# Accelerating Variance Reduction: Optimal Complexity of Finite-Sum Optimization

Project proposal, Statistical Learning Theory

Beline Yuan Paul Ruiz Alliata Louis Darrigol

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## 1 Motivation

Variance-reduced methods such as SAGA and SVRG solve finite-sum problems at the cost of only a few passes over the data, while Nesterov acceleration improves the dependence of full-batch gradient descent on the condition number $\kappa$. Chapter 5 of Prof. Bach's *Learning Theory from First Principles* shows that each method wins in a different regime. Counting individual gradient evaluations, SAGA costs $O\big((n+\kappa)\log(1/\varepsilon)\big)$, whereas accelerated full-batch gradient descent costs $O\big(n\sqrt{\kappa}\,\log(1/\varepsilon)\big)$. This raises two questions. First, can one get the best of both worlds? Second, which steps of variance-reduced algorithms explain the gap between the predicted upper bounds and the observed behavior, and for which problem geometries are these bounds particularly conservative? Answering them connects the theory of Chapter 5 directly to the practical behavior of variance-reduced optimization.

## 2 Problem Setup

Consider finite-sum objectives

$$F(\theta) = \frac{1}{n}\sum_{i=1}^{n} f_i(\theta),$$

where each $f_i$ is smooth and $F$ is strongly convex. Our main experimental setting would be $\ell_2$-regularized logistic regression, because it satisfies the assumptions used in the SAGA analysis while remaining simple enough that the relevant constants and optimum can be accurately estimated.

We would run SAGA with the update studied in Section 5.4.4 and track both parameter convergence and the gradient-memory error. The primary Lyapunov quantity would mirror the one appearing in Proposition 5.9:

$$V_t = \|\theta_t - \theta^\star\|^2 + \Delta \sum_{i=1}^{n} \|z_i^{(t)} - \nabla f_i(\theta^\star)\|^2.$$

We would generate synthetic datasets where we can independently vary dataset size $n$, regularization strength, and conditioning of the feature covariance matrix. This gives control over the condition number and allows us to study when the theoretical assumptions become loose.

# 3 Hypothesis A

*Can variance reduction be applied to Nesterov acceleration, provided the momentum is corrected.* More precisely:

(A1) **Lower bound.** Any span-based method with i.i.d. uniform sampling needs $\Omega\big((n+\sqrt{n\kappa})\log(1/\varepsilon)\big)$ gradients. For an arbitrary index sequence, the $\sqrt{n\kappa}$ term survives but the $n\log(1/\varepsilon)$ term weakens by a factor $\log(n/\kappa)$ when $\kappa \ll n$.

(A2) **Naive acceleration.** Plugging the SVRG estimator into Nesterov's algorithm.

(A3) **Corrected acceleration works.** Adding a "negative momentum" term toward a reference point (loopless Katyusha) makes the variance absorbable, and yields the optimal rate. We will reproduce this proof in the Lyapunov style of the notes.

(A4) **Empirical exponents.** On the hard instance of (A1), the number of gradients to reach a fixed accuracy grows like $\kappa^1$ for SAGA and $\kappa^{1/2}$ for L-Katyusha, in the regime $\kappa \gg n$.

# 4 Hypothesis B

*Could the variance reduction algorithm be optimized directly by itself?* More precisely:

(B1) **Quadratic modification.** We will try to change the definition of the variance reduction algorithm by adding a quadratic factor to the recursion defining the gradient descent step, and see which steps of the proof are too loose:

$$
\begin{aligned}
g_t &= f'_{i(t)}(\theta_{t-1}) + \Big(\frac{1}{n}\sum_{i=1}^{n} z_i^{(t-1)}\Big) - z_{i(t)}^{(t-1)},\\
\theta_t &= \arg\min_{\theta}\ \Big\{\langle g_t, \theta\rangle + \frac{1}{2\gamma}\,(\theta-\theta_{t-1})^\top P\,(\theta-\theta_{t-1})\Big\} = \theta_{t-1} - \gamma\,P^{-1} g_t,\\
z_{i(t)}^{(t)} &= f'_{i(t)}(\theta_{t-1}), \qquad z_j^{(t)} = z_j^{(t-1)} \text{ for } j \neq i(t).
\end{aligned}
$$

Here, $i(t) \in \{1,\dots,n\}$ is an index drawn uniformly at random at iteration $t$, $f'_i = \nabla f_i$, and $z_i^{(t)} \in \mathbb{R}^d$ is the stored gradient of $f_i$ (the gradient memory), initialized at some $z_i^{(0)}$. The vector $g_t$ is the variance-reduced gradient estimator, which is unbiased: $\mathbb{E}[g_t \mid \theta_{t-1}] = \nabla F(\theta_{t-1})$. Finally, $\gamma > 0$ is the step size and $P \in \mathbb{R}^{d\times d}$ is a symmetric positive definite matrix defining the quadratic metric; $P = I$ recovers standard SAGA.

(B2) **Trajectory-dependent constants.** We will try to identify whether trajectory-dependent local quantities in the asymptotic analysis describe the convergence better than the global worst-case constants used in the theorem. In particular, we hypothesize that quantities such as the effective curvature experienced along the optimization trajectory may be significantly tighter than the global strong-convexity constant. As a result, the strong-convexity step or another specific inequality in the proof may explain most of the difference between the theoretical and empirical contraction rates.

(B3) **Experimental diagnosis.** We will run SAGA for different curvatures and compare the actual experimental values with the upper bound to see where the loose parts of the proof are. If (B1) is successful, we will add a run with the quadratic factor.

The significance is that the project would not simply show that the theorem is conservative; it would identify why it is conservative and how this depends on problem conditioning. We may then be able to propose another version of the theorem with a tighter evolving constant, but probably more computationally expensive.

# 5 Experimental Plan

The first stage would be to reproduce the SAGA algorithm and Proposition 5.9 computationally. We would solve each logistic-regression problem to high precision to obtain an approximation of $\theta^\star$, then run many SAGA trajectories with different random seeds. We would similarly instrument the square-norm inequality, the cocoercivity bound, and the variance-to-second-moment inequality.

The experimental pipeline would be approximately:

1. Generate strongly convex logistic-regression problems with controlled $n$, condition number, and covariance anisotropy.
2. Compute or accurately approximate $\theta^\star$.
3. Run SAGA over multiple random seeds.
4. Record $V_t$, actual contraction rates, gradient variance, and the quantities appearing at every step of the proof.
5. Compute a "tightness ratio" for each inequality.
6. Compare these ratios across problem geometries.
7. Test whether replacing global quantities such as $\mu$ with trajectory-dependent effective quantities better explains the observed rate.

A possible intermediate theoretical result would be to define an effective curvature

$$\mu_{\text{eff}}(t) = \frac{(\theta_t - \theta^\star)^\top \nabla F(\theta_t)}{\|\theta_t - \theta^\star\|^2}$$

and establish basic relations such as

$$\mu \leq \mu_{\text{eff}}(t).$$

We could then test whether replacing $\mu$ by empirical values of $\mu_{\text{eff}}$ accounts for a significant fraction of the theory–practice gap. Another possible extension would be optimizing the Lyapunov coefficient $\Delta$ or step size $\gamma$ within the same proof framework instead of using the convenient constants selected in the textbook.

**Expected outputs.** The key results would likely be three figures. One compares empirical convergence against the Proposition 5.9 bound. A second shows the tightness of each proof inequality during training. A third varies conditioning or feature anisotropy and shows which proof step becomes the main bottleneck.

# 6 Use of generative AI

We used a large language model (Claude) for formatting and to double-check the feasibility of our ideas. Representative prompts:

- *"Transform this into LaTeX."*

- *"Can you tell me whether this is a relevant idea, not so much in terms of the method's success but rather in terms of the mathematical depth of the problem: will the associated mathematics be interesting?"*
- *"What has already been done on variance reduction?"*